

# How Our Best Theories Coarse-Grain Reality: One Field, Its Shadows, and What Each Blur Throws Away

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## Preamble — what this article does NOT claim

1. **No new results.** Every physics claim below is a standing corpus result, cited to its paper; this article only arranges them under one lens.
2. **The lens is a reading, not a theorem.** “Kept-column  $\rightarrow$  variables, discarded-column  $\rightarrow$  symmetries” is an organizing principle that the corpus results instantiate; it is not itself derived from the primitives.
3. **Tiers vary by entry and are marked.** One entry is build-certified; most are structural (conditional on ED’s postulates); two are schematic. ED’s results are conditional derivations from thirteen posited primitives — never derivations from nothing — and the framework is not experimentally confirmed.
4. **The full Einstein field equations are not derived** (weak field and khronometric class only), and the Born-rule account is a reconstruction with a named residual. The honest edges are flagged where they occur.

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## 1. What Event Density is, in one page

Event Density is an account of physics built on a **discrete, relational substrate**: a participation graph in which chains participate in channels. Each channel carries a **bandwidth**  $b \geq 0$  and a phase, combined in the participation amplitude  $P = \sqrt{b \cdot e^{i\pi}}$ . There is one process primitive: **commitment** (P11) — the irreversible act by which an uncommitted participation becomes a definite, recorded fact. The arrow of time is not a boundary condition on top of reversible laws; it is this act, written into the law. Everything else — space, geometry, matter, fields, the quantum formalism — is developed as structure in how committed events relate, as conditional derivations from thirteen posited primitives, with the form/value split always named (*form-forced, value-inherited*).

ED does not overturn any proven physics. General relativity and quantum mechanics remain as successful as ever; the claim of this article is about *where they sit*: each is what the substrate looks like after a particular blur. This article introduces **no new derivation** — it arranges standing corpus results under a single lens, at their honest tiers. The corpus lives at [github.com/allen-proxmire/ED-generative](https://github.com/allen-proxmire/ED-generative) (DOI 10.5281/zenodo.20149496); its front door is the Report, *Event Density: A Unified Framework for Physics*, which states every result below with the papers behind it. The two load-bearing

sources for this article are the curvature-emergence line (Paper\_MetricFromTheGraph, Paper\_027, GR-I.IV, Paper\_QuadraticStrain) and the coarse-graining line (Paper\_Continuum\_KineticLatticeGas and the two-layer results).

## 2. The two organizing facts

### 2.1 One field, two shadows

In ED, **matter and geometry are two readings of the same field**. A mass is a *depletion* of bandwidth —  $b$  trapped in a bound, self-sustaining loop (“mass without mass”). The metric is the *reading* of that same field’s connectivity —  $g \sim 1/b$ , distances read off the graph, never assigned (Paper\_MetricFromTheGraph). Neither matter nor geometry is a substrate primitive; both are coarse-grained shadows of the one bandwidth field. Everything in this article happens at, or above, the **seam** where those two shadows meet.

### 2.2 The ledger: what is kept becomes the variables; what is discarded becomes the symmetries

A coarse-graining is an average, and **averages are permutation-invariant: a sum does not remember the order of its terms**. So when many commitments are blurred into smooth variables, the *order of events* is not stored anywhere in the output. A law written purely in the blurred variables therefore *cannot* contain an arrow — not because reality is reversible, but because the theory’s vocabulary no longer has a word for “which way.” Its reversibility is **ignorance of order, formalized**.

This generalizes. Every blur keeps a two-column ledger:

- **KEPT** (smoothed)  $\rightarrow$  the theory’s *variables*;
- **DISCARDED**  $\rightarrow$  the theory’s *symmetries*.

**Emergent symmetry is forgotten information**. Reversibility, unitarity, frame-democracy, instantaneous action, the continuous field — each celebrated symmetry below is the receipt for a specific discard. A pointillist canvas makes the principle visible: the finished scene contains nothing that records which dot was placed first, so the *canvas* is “reversible” even though the *painting of it* was not.

*Scope, stated precisely: the order-erasure mechanism applies to blurs whose kept variables are order-symmetric averages — which is the situation for the geometric and unitary theories below. A blur that keeps fluxes or currents, or that tracks the discard itself, inherits a direction — which is exactly how diffusion (§11) and thermodynamics (§13) retain the arrow the others lost. The ledger’s claim is not that every coarse-grained law is reversible; it is that each theory’s symmetries are read off its discard column.*

## 3. G — the conversion factor at the seam (*structural*)

Newton’s constant is the cleanest first example, because  $G$  is the exchange rate between **two separately coarse-grained things**:

1. **Mass** — bound bandwidth. Already an emergent, coarse-grained reading of the substrate, not a primitive.
2. **Curvature** — the emergent metric  $g \sim 1/b$ . Also a shadow, read off connectivity.

$G$ ’s whole job is converting one into the other ( $a = GM/R^2$ ). It lives in neither coarse-graining; it lives **at the seam between the two**, a conversion factor between two shadows. That is why it is doubly downstream — both blurs must have already happened before  $G$  is even meaningful — and why “least fundamental constant” and “first to emerge” turn out to be one fact:  $G$  is assembled entirely from the

starting units ( $G = c^3 \ell_P^2 / \hbar$ , Paper\_027), appearing at the first place the two shadows meet. Consistent with this,  $G$  appears nowhere in physics outside gravity — no  $G$  in the entire Standard Model — exactly what one expects of a constant that belongs to a seam rather than to the substrate.

#### 4. General relativity — the law of the seam (*structural; weak field closed, full EFE open*)

Where  $G$  is the exchange *rate* at the seam,  $GR$  is the exchange *law*. Einstein’s equation has geometry on the left and matter on the right,  $G_{\mu\nu} = 8\pi G \cdot T_{\mu\nu}$ , and in ED both sides are coarse-grainings of the one bandwidth field — plus one thing the blur removed entirely:

1. **The left side (geometry) is itself a double coarse-graining.** The metric is read off graph connectivity, and that takes two steps: the raw hop-count metric is faceted and direction-dependent (layer 1, the ballistic reading); only further averaging produces isotropy (layer 2 — *measured* in the Paper\_MetricFromTheGraph probes: anisotropy  $0.35 \rightarrow 0.00$  as the reach coarsens), while the clean  $g \sim 1/b$  exponent is *derived from the holographic channel count*, not from transport — its direct layer-2 transport confirmation is that paper’s named open item. The smooth Lorentzian manifold sits two floors above the substrate.
2. **The right side (matter) is the other coarse-graining.** Stress-energy is the bookkeeping of participation content — mass as trapped  $b$ , energy flux as moving  $b$ . The same field, read as “stuff” instead of “shape.”
3. **And the arrow was not coarse-grained — it was coarse-grained away.**  $GR$ ’s spacetime is a block: reversible, no preferred frame, diffeomorphism-invariant. The substrate has a preferred direction (the khronon — the arrow made dynamical,  $GR-II$ ), and the averaging blurs it to near-invisibility, by the ledger mechanism of §2.2.  $GR$ ’s celebrated reversibility is not a deep fact about reality; it is the artifact of a blur that erased the *order* of commitment. This is why time is missing-or-awkward in  $GR$  — the problem of time: it was not left out, it was averaged out. The residue the blur did not quite erase is precisely ED’s khronometric departure from pure  $GR$  — the preferred-frame parameters  $\alpha_1, \alpha_2$ .  $GR-IV$  computes them:  **$\alpha_2$  vanishes exactly** (a structural consequence, independent of tuning), and  **$\alpha_1$  is suppressed some seventy orders of magnitude below current bounds** — in that paper’s own words, *the arrow is real, but it leaves no shadow*. The residue is distinctive in principle and unmeasurably small in practice.

**The payoff.** The century-old brute miracle of  $GR$  — *why* matter and geometry are locked together at all — dissolves at the substrate: **they were never two things**. Mass is a depletion of  $b$ ; curvature is the reading of  $1/b$ . One field, two shadows; Einstein’s equation is the coarse-grained **consistency condition that the two readings of the same field agree** — the law of the seam — and  $G$  is the unit conversion that law uses. The features  $GR$  is most celebrated for (background independence, no preferred frame, timeless elegance) are, on this reading, the signature of *how much averaging happened*, not glimpses of the bottom.

*Honest tier:* ED delivers the weak field ( $GR-I$ ) and the khronometric class with its dynamical rule ( $GR-II/III$ ); the full nonlinear EFE is not derived.

#### 5. Special relativity — keep the cone, blur the frame (*structural ingredients, lens-read*)

**Kept:** one propagation speed. The substrate’s retarded kernel fixes a single rate of becoming,  $c$  — the tick-to-hop conversion — and with it a causal cone. **Blurred:** the grain, into a continuum. **Discarded:** the substrate’s preferred simultaneity (the khronon’s foliation). **The artifact:** the relativity of simultaneity, and the celebrated democracy of observers — “no preferred frame.” In ED the frame exists

and *hides*; the same residue pair  $\alpha_1, \alpha_2$  measures how well it hides. SR's most philosophically radical feature is, on this ledger, an amnesia: the substrate knows which slicing is real, and the blurred theory has forgotten.

## 6. Newton — blur twice more (*structural + standard limit*)

From GR, Newtonian gravity discards two further things. **Discard retardation** (the finite  $c$  of propagation) and the artifact is **instantaneous action at a distance** — the very feature Newton himself refused to defend (“hypotheses non fingo”); his discomfort was, on this reading, well-placed: instantaneity is the scar of a discarded delay. **Discard the horizon cross-term** (the interference contribution from the cosmic decoupling surface, Paper\_QuadraticStrain) and the artifact is the *clean, universal* inverse-square law. An honesty pin: Newton-from-GR is technically a *limit* (weak field, slow speeds) rather than a statistical average — kin to a coarse-graining, but a slightly different step; the discarded-column logic applies to both.

## 7. Schrödinger — the window between commitments (*structural: the QM-emergence arc*)

Unitary quantum mechanics is produced by a different operation: not an average over events but a **restriction of attention** — watch only the stretches *between* commitments. **Kept:** the amplitude and phase structure,  $P = \sqrt{b \cdot e^{i\pi}}$ , evolving coherently. **Discarded:** the commitments themselves — windowed out of the description. **The artifact: unitarity.** QM's exact reversibility is the same amnesia as GR's, achieved by other means: *GR averaged the order out of many events; Schrödinger windowed the events out entirely.* Two theories, one discard — which is why *both* pillars of modern physics lack the arrow, why both have a “problem of time,” and why a century of tying them *to each other* has failed. They are two blurs of one substrate; the join runs through the thing they both discarded. One cannot quantize GR — but one can discretize both. (*The identification of the two discards as one primitive is the Report §6 unification move, which carries its own guardrail: a structural synthesis, not a theorem, with no delivered prediction yet.*)

## 8. Born and collapse — the part no blur could digest (*reconstruction tier; Report §4*)

The Born rule and the collapse are **not a coarse-graining at all**. They are the substrate showing through: the commitment itself — irreversible, stochastic, basis-selecting — surviving into the blurred theory *unblurred*. This is why textbook quantum mechanics has its notorious two-part structure: smooth deterministic evolution punctuated by abrupt probabilistic collapse. On the ledger the mystery inverts: the smooth part is the *artifact* (the between-commitments window), and the “bolted-on” collapse postulate is the *one genuine piece of the substrate* in the theory. The measurement problem is what it feels like to mistake the windowed theory for the whole — to treat the artifact as fundamental and the substrate's signature as an embarrassment. (*Tier: the ED measurement account is a reconstruction — commitment selects the pointer basis, Born non-contextuality grounded — with the Solèr-rigor residual named in the Report.*)

## 9. Classical mechanics — commit constantly (*schematic*)

Blur once more: let commitment happen so frequently, at such multiplicity, that no phase coherence survives between events. **Discarded:** the phase. **The artifact:** definite trajectories and effective deter-

minism. The classical world is the commit-saturated limit of the window of §7 — the regime in which the substrate’s bookkeeping and its record-keeping coincide.

## 10. Maxwell — keep the coherent component (*structural; closed*)

The electromagnetic field is what survives when the polarity transport of many channels is blurred and split into a coherent component and disorder: **kept**, the coherent piece, whose limit is Coulomb; **discarded**, the per-channel phase granularity. **The artifact: the smooth, continuous field itself.** There are no fields at the bottom of ED; “field” is the name of this shadow — which is why the corpus flags any treatment of fields as primitives as a category error.

## 11. Transport and diffusion — the certified rung (*layer 1 certified; the layer-2 step identified*)

One entry anchors the whole ledger in arithmetic rather than metaphor, because the blur has been *run*, end to end, on the certified substrate. Coarse-graining the commit-dynamics directly yields a **ballistic kinetic lattice-gas** (layer 1) — certified by build, not asserted. The further blur to the **diffusion PDE** (layer 2) proceeds by decorrelation: **discard velocity memory**, and the artifact is **Markov memorylessness** — the defining “symmetry” of diffusion, its indifference to history. And the decorrelation step is not merely identified-and-pending — it was *tested from the ED side, with a negative result*: the pure substrate **refuses** it (persistence  $\sim 0.45$ , sub-diffusive; the PDE is reached only by leaving ED). Far from undermining the ledger, that supports it: the second blur is the *describer’s* forgetting, not the substrate’s dynamics — the diffusion equation is a statement about our bookkeeping, and the substrate declines to do our forgetting for us. This section is the existence proof: “take the substrate, blur, obtain the continuum law” is a computation ED has actually performed at least once.

## 12. MOND — where a blur fails (*structural, conditional on P-Quadratic-Strain and the P12-Coh operationalization*)

The ledger also *locates* breakdowns. Newton (§6) discarded the horizon interference term. That discard is *safe* where the local field dominates — and **unsafe below  $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ , where the discarded term is the larger one** ( $a = a_N + \sqrt{a_N a_0}$ ), Paper\_QuadraticStrain; the constructive sign the cross-term needs — once that paper’s one named residual — was subsequently *supplied*, build-verified at MEASURED tier in Paper\_PhaseCoherence\_P12Coh conditional on the P12-Coh operationalization, so it is no longer an open assumption). Galactic rotation curves are not evidence of a new substance; they are **the place where Newton’s blur stops being a good approximation** — the un-blurrrable residue, surfacing exactly where the corpus result puts it, and just where the ledger’s reading expects a discarded column to resurface. MOND phenomenology is a failed coarse-graining diagnosed, not a force invented.

## 13. The closer: thermodynamics — the receipt (*interpretive, grounded*)

Every theory above became elegant by forgetting something. **Thermodynamics is the one continuum theory that kept an arrow — because thermodynamics is the accounting of the discard pile itself.** Entropy *counts what the blurs threw away*; the second law is the statement that the pile only grows. It is the receipt for the coarse-graining — the invoice every other theory misplaced.

And, on this ledger, it dissolves the oldest paradox in the subject. Loschmidt’s objection — *how can irreversible thermodynamics emerge from reversible microscopic laws?* — assumes the reversible laws

are the bottom. On this ledger they are not: **the substrate was never reversible; reversibility was the artifact**, manufactured by blurs that discarded the order of commitment. Irreversibility does not need to be smuggled back in through special initial conditions. It was there all along; the “fundamental” reversible laws are the amnesiacs, and thermodynamics is the one department that kept the records.

#### 14. The recipe table

| Get                        | Keep                                                   | Blur / discard                                                                                                  | The artifact-symmetry it buys                                                      | Tier                     |
|----------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------|
| <b>Schrödinger</b>         | the amplitude<br>$\sqrt{b \cdot e^{i\pi}}$             | window between commitments; discard the commitments                                                             | unitarity (reversibility)                                                          | structural               |
| <b>Born / collapse</b>     | —                                                      | <i>not a blur</i> : the commitment surviving unblurred                                                          | none — it <i>breaks</i> the artifact-symmetries; QM’s two-part structure explained | reconstruction           |
| <b>Classical mechanics</b> | expectation values                                     | commit-often limit; discard phase                                                                               | definite trajectories, determinism                                                 | schematic                |
| <b>SR / Minkowski</b>      | one propagation speed (the cone)                       | blur the grain; discard the khronon frame                                                                       | no preferred frame; relativity of simultaneity                                     | structural               |
| <b>GR</b>                  | geometry- and matter-readings of $b$                   | blur graph $\rightarrow$ metric ( $\times 2$ ) and bound- $b \rightarrow T_{\mu\nu}$ ; discard commitment order | reversibility + diffeo invariance (residue $\alpha_1, \alpha_2$ )                  | structural; EFE open     |
| <b>Newton</b>              | GR’s diagonal term                                     | discard retardation and the horizon cross-term                                                                  | instantaneous action; clean inverse-square                                         | structural + limit       |
| <b>Maxwell</b>             | the coherent component of polarity transport densities | discard per-channel phase granularity                                                                           | the smooth continuous field                                                        | structural; closed       |
| <b>Diffusion</b>           |                                                        | layer-1 ballistic $\rightarrow$ layer-2; discard velocity memory                                                | Markov memorylessness                                                              | <b>layer 1 certified</b> |

| Get                   | Keep                    | Blur / discard                                                             | The artifact-symmetry it buys                                      | Tier                    |
|-----------------------|-------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------|
| <b>MOND</b>           | —                       | <i>the anti-recipe:</i><br>where Newton's discard dominates (below $a_0$ ) | none — the discarded column resurfacing                            | structural, conditional |
| <b>Thermodynamics</b> | the discard pile itself | —                                                                          | the one law that <i>keeps</i> the arrow; the 2nd law = the receipt | interpretive, grounded  |

## 15. What the lens buys

**It explains the shape of the canon.** Why do the two pillars of physics share the same two embarrassments — a missing arrow of time and an unexplained measurement event? Because they are two blurs of one substrate that discarded the same thing, one by averaging it out (GR) and one by windowing it out (Schrödinger). And why has a century of joining them theory-to-theory failed? Because hands do not attach to hands; they attach through the body. The unification move is not to quantize gravity — it is to **discretize both**, re-attaching each blur to the substrate it came from.

**It explains why each theory is so good.** A blur is *exact about what it keeps*. Inside its kept-column, each theory is unbeatable, and ED predicts no deviation there — which is why ED overturns nothing.

**It locates the predictions.** A discarded column never fully vanishes; it *leaks* — though not every leak is measurable. The arrow's leak into gravity ( $\alpha_1, \alpha_2$ ) is computed and real but suppressed beyond any current experiment (GR-IV). The live tests are the leaks that *are* reachable: the sub- $a_0$  regime (Newton's discard resurfacing in galaxies), and the quantum-to-classical boundary at internal multiplicity  $M_{\text{crit}}$ , with the pre-registered 140–250 kDa matter-wave crossing (the commitment leaking into interferometry). ED's falsifiable content is, almost without exception, **the discarded columns leaking back** — which is exactly where a substrate theory *should* differ from its own shadows.

**Honest close.** The lens is only as strong as the results it organizes: one rung is certified, most are structural and conditional on the thirteen primitives, two are schematic, the full EFE and the dimensionless constants are inherited or open, and no argument-ending prediction is confirmed. What the ledger offers is not proof but *shape*: one substrate, two shadows, a canon of blurs — and every famous symmetry a receipt.

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## Cross-references

- **Substrate & primitives:** physics-papers/foundations/Paper\_087\_13Primitives.md; the Report (repo root).
- **Geometry emergence (×2 layers):** physics-papers/gravity/Paper\_MetricFromTheGraph\_ForcedTo3D.md; GR-I..IV.
- **The seam:** Paper\_027\_Newtons\_G.md ( $G = c^3 \ell_P^2 / \hbar$ ); Paper\_QuadraticStrain\_v1.md (diagonal = Newton, off-diagonal = MOND); Paper\_029\_a0.md (the horizon term); Paper\_PhaseCoherence\_P12Coh.md (the constructive interference sign, supplied at measured

tier).

- **QM window & commitment:** the QM-emergence arc (physics-papers/qm-kinematics/), Report §4; the Lindblad derivation (working repo, arcs/arc-Q/lindblad\_extension.md).
- **Certified coarse-graining:** Paper\_Continuum\_KineticLatticeGas and the two-layer coarse-graining results.
- **Maxwell as coherent shadow:** the CG-decomposition papers (substrate-evaluation).
- **The M\_crit leak:** physics-papers/q-compute/Paper\_060\_Mcrit\_Unification.md; physics-papers/predictions/QC-Mass-Extrapolation\_InProcess/.